

Data Center Failure Prediction System

Run Guide & Paper Methodology Overview (Modules 1 & 2 + 5-Point Leakage Audit)

1. What is this Project? (Simple Explanation)

Imagine a massive cloud server infrastructure running thousands of tasks 24/7. Before a machine fails, it exhibits subtle **telemetry degradation patterns** — such as CPU usage drift, memory exhaustion, disk I/O bottlenecks, or scheduler latency spikes.

This project builds an **Artificial Intelligence (AI) Precursor Warning System using XGBoost** that detects failure precursors **30 minutes BEFORE a server crash occurs**, allowing live task migration and zero downtime.

2. How to Run the Project (Step-by-Step)

You can execute the entire pipeline in **one step** from your terminal:

Step 1: Open Command Prompt or PowerShell.

Step 2: Navigate to your project folder:

```
cd "c:\Users\HINATA\Music\Bakya Akka"
```

Step 3: Execute the master pipeline script:

```
python run_pipeline.py
```

3. How the Pipeline Works (The 2 Modules & 5-Point Audit)

The pipeline is divided into two modules with strict temporal isolation and **zero data leakage**:

Module & Component	Methodology & Workflow Implementation
Module 1 Module1_...ipynb	Stochastic Telemetry Simulation: Generates 50,000 continuous time-series rows across 25 machines using Ornstein-Uhlenbeck processes superposed with non-linear degradation drift. Backward-Only Window Features: Computes 52 rolling mean, std, trend, and CV features over $W \in \{5, 15, 60\}$ timesteps. Strict History Rule: Enforces <code>min_periods=W</code> so features rely strictly on past raw telemetry ($t' \leq t$).
Module 2 Module2_...ipynb	Forward Horizon Labeling ($SH=30m$): Labels $y=1$ for timesteps in $[f_{idx}-H, f_{idx})$. Chronological Split with Embargo Purging: 70% Train, 15% Val, 15% Test per machine. Purges $W_{max}-1$ feature lookback rows and SH horizon lookahead rows at split boundaries. Full 5-Point Leakage Audit: Verifies Feature Bounds, Target Isolation, Temporal Separation, Feature Embargo Isolation, and Target Horizon Isolation. Forward-Chaining CV & Hyperparameter Tuning: Tunes XGBoost hyperparameters using expanding temporal folds before final model fitting.

4. Benchmark Performance Results

Final benchmark evaluation on the out-of-time test partition after embargo purging and forward-chaining CV:

Model	Threshold	Precision	Recall	F1-Score	PR-AUC	ROC-AUC
Proposed XGBoost	0.730	55.4%	100.0%	71.3%	0.838	0.999
Random Forest Baseline	0.480	51.4%	100.0%	67.9%	0.463	0.999
SVM Baseline	0.330	64.7%	91.7%	75.9%	0.849	0.999

Key Methodological Achievements:

- **100% Zero Leakage:** Passed all 5 programmatic audit checks including boundary embargo isolation.
- **High Precursor Recall:** Proposed XGBoost detects failure precursors with **100% Recall** on out-of-time test data.
- **30-Minute Early Warning:** Achieves a mean precursor lead time of **30.0 minutes** prior to failure.

5. Generated Artifacts & Directory Structure

All generated output artifacts are organized in dedicated output directories:

- module1_outputs/: Raw telemetry & backward sliding-window feature Parquet files.
- module2_outputs/:
 - xgboost_m1_precursor_model.json (Trained XGBoost model)
 - m1_benchmark_results.csv (Benchmark evaluation table)
 - m1_performance_and_feature_importance.png (PR curves & Feature importances chart)
 - m1_lead_time_distribution.png (Precursor lead-time distribution chart)